

Confidence-Aware Edge Routing for Robotic TCM Tongue Imaging: Edge-IQA and LangGraph Closed-Loop Acquisition

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Abstract

Robotic acquisition of tongue images for traditional Chinese medicine (TCM) tele-diagnosis requires tight coupling between motion, on-device quality assessment, and upload policy under variable network delay. We present a cloud–edge–device stack that keeps learning-based perception out of the 1 kHz ROS control loop: a lightweight *Edge-IQA* scores each frame on the gateway, and a *LangGraph* router decides among cloud upload, edge resampling, or fail-safe abort within a bounded retry budget. On the ShezhenV3-COCO test split (553 images, real Edge-IQA scores, 500 frames $\times$ 3 seeds, baselines B0–B3), the proposed B2 configuration reduces median end-to-end RTT (M1 p50) from 383.6 ms to 341.2 ms versus naive upload (B0) while raising valid-frame rate (M2) from 0.52 to 1.00. Edge-IQA is calibrated on 572 validation pairs ($\tau=0.498$) and runs at p95 ≈ 9.6 ms per frame on CPU. An auxiliary Franka closed-loop track (50 trials) confirms routing decisions are reproducible on the execution stack.

1 Introduction

Tele-diagnosis for tongue inspection in traditional Chinese medicine (TCM) needs repeatable, high-quality images from a manipulator-mounted camera before cloud analytics and electronic medical records (EMR) integration [6, 8]. Clinical tongue diagnosis relies on color, texture, coating, and shape cues that degrade under motion blur, uneven lighting, and incorrect working distance [8]. Naive pipelines upload every frame to the cloud, wasting bandwidth on blurred or poorly exposed captures and inflating end-to-end latency when the link is slow.

Motivation. Cloud–edge collaborative robotics promises to offload heavy vision models while keeping acquisition loops responsive. However, *garbage-in garbage-out* (GIGO) effects are acute in

TCM imaging: a single blurred upload triggers wasted round-trip time (RTT) and may propagate low-confidence features into downstream classifiers. Edge-side quality gating must therefore be (i) fast enough for closed-loop capture, (ii) interpretable for audit, and (iii) calibratable on real tongue corpora rather than synthetic placeholders alone.

Approach. We study a **cloud–edge–device** architecture for a Franka Research 3 (FR3) platform with ROS 2 and MoveIt motion execution. Perception and routing policies live in a Python edge gateway (`franka_api_server`) and in the `doctor/paper1` research tree—not inside hardware plugins—so algorithms remain testable without a robot. A lightweight *Edge-IQA* module fuses Laplacian sharpness and exposure into $Q_{\text{img}} \in [0, 1]$; a LangGraph router decides `upload_cloud`, `resample_edge`, or `fail_safe` under retry budget K .

Full-dataset evaluation. After phase-2 synthetic prototyping, we validate on the public **ShezhenV3-COCO** dataset (6,719 images): τ is calibrated on 572 validation pairs (clear + synthetic blur), and the main matrix replays 500 test frames per seed with *real* Edge-IQA scores from 553 held-out images.

Contributions.

1. **Edge-IQA:** an $O(HW)$ quality index fusing Laplacian sharpness and exposure, calibrated at $\tau^*=0.498$ on ShezhenV3 validation with CPU p95 below 10 ms at 512 px.
2. **Confidence-aware LangGraph routing:** a three-branch policy (`upload_cloud`, `resample_edge`, `fail_safe`) with retry budget $K=2$, logged as JSONL for reproducibility.
3. **Dual-track evaluation:** baselines B0–B3 on real TCM test scores supply Table 9; an auxiliary Franka track (M6) validates execution-

stack trends without mixing CSVs into primary statistics.

Paper organization. Section 2 reviews medical robotics, IQA, and edge orchestration. Section 3 presents the system architecture, Edge-IQA, routing, and closed-loop protocol. Section 5 describes the ShezhenV3 cohort, calibration, main matrix, and ablations. Section 8 concludes with limitations and future work.

2 Related Work

Robotic medical imaging. Surgical and diagnostic robots increasingly use ROS 2 and MoveIt for motion planning [1, 4]. Tele-operated and autonomous cameras must respect joint limits, collision avoidance, and repeatable viewpoints—requirements that motivate *skills* (calibrated poses) rather than ad-hoc joint teleoperation. We adopt named skills (`go_to_tongue_pose`, `go_to_face_pose`) instead of online impedance tuning for acquisition poses, matching clinical repeatability requirements while keeping Paper I independent of force-control research threads.

Image quality assessment. **Full-reference** metrics (PSNR, SSIM, MS-SSIM) compare captured frames to pristine references and remain the gold standard for compression benchmarking, but are inapplicable when each patient acquisition lacks a reference image. **Hand-crafted NR-IQA** (Laplacian variance, Tenengrad, BRISQUE, NIQE, PIQE) operates without references; BRISQUE/NIQE model natural-scene statistics [5] and correlate with human opinion on in-the-wild photos, yet their scores require additional mapping before threshold routing and seldom expose blur/exposure semantics needed for clinical audit logs. **Deep NR-IQA** (DB-CNN [7], NIMA, HyperIQA, MUSIQ) improves correlation on benchmark databases by learning perceptual features, at the cost of GPU inference, model versioning, and limited interpretability when a frame is rejected.

Edge deployment in closed-loop robotics favors linear-time operators with auditable failure modes; our Edge-IQA follows this line with explicit blur/exposure flags and a calibrated τ for LangGraph routing (Table 2). We do not claim mean-opinion-score (MOS) prediction superiority on

generic IQA leaderboards; the goal is *edge-side acquisition gating* under CPU latency and reproducibility constraints on TCM tongue imagery.

Edge–cloud orchestration. Split inference and early-exit policies reduce bandwidth in mobile vision [2, 3]. Neurosurgeon-style frameworks partition DNN layers across device, edge, and cloud; Paper I differs by gating *upload decisions* before any heavy cloud model executes, using a lightweight scorer rather than partial DNN migration. LangGraph-style state machines express branching policies with auditable logs, complementary to monolithic ROS nodes that embed if-else logic without structured trace export.

TCM tongue analysis. Vision pipelines for tongue classification, segmentation, and color analysis are mature in the cloud [6, 8]. Public corpora such as ShezhenV3-COCO provide bounding boxes and condition labels for *downstream* diagnosis models. Paper I focuses on *acquisition quality and routing* upstream of these models: even state-of-the-art classifiers degrade when motion blur or exposure shifts dominate input pixels. Our full-dataset evaluation (6,719 images) connects Edge-IQA calibration to this community resource rather than synthetic placeholders alone.

Positioning summary. Compared with generic edge offloading, we target robotic *capture* loops; compared with NR-IQA literature, we prioritize thresholdable Q_{img} and JSONL auditability; compared with TCM diagnosis papers, we address frame usability before classification.

3 Method

4 System Architecture

We target a cloud–edge–device stack for robotic tongue-image acquisition in traditional Chinese medicine (TCM) tele-diagnosis. The design separates *real-time* acquisition and quality control at the edge from *offline* or high-latency cloud analytics, while keeping algorithm modules testable outside the robot operating system (ROS) runtime.

4.1 Device Layer

The device layer consists of a Franka Research 3 (FR3) manipulator, RGB imaging, and ROS 2-based motion execution. MoveIt provides point-to-point (PTP) joint-space motions. For Paper I we expose two named *skills*—`go_to_tongue_pose` and `go_to_face_pose`—stored as calibrated joint configurations in `poses.yaml` and invoked through the edge gateway. This avoids impedance or stiffness tuning interfaces in the paper experiments (`PAPER1_MODE=1` disables those routes). Simulated and fake-hardware stacks (MoveIt fake controller, optional Gazebo camera) support reproducible auxiliary trials without a physical robot.

4.2 Edge Layer

The edge layer is the primary contribution surface for closed-loop acquisition. A FastAPI server (`franka_api_server`) acts as the edge gateway on port 8000, bridging HTTP clients to ROS 2 actions for motion and error recovery. **Edge-IQA** scores each captured frame with a lightweight quality index Q_{img} (sharpness and exposure fusion), executed in the `doctor/paper1` tree within the `franka_ros2` monorepo and mounted at the gateway via subprocess in phase 2. **LangGraph** implements confidence-aware routing over a small state machine: `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route`, with branches `upload_cloud`, `resample_edge`, and `fail_safe`. The HTTP client `sim/franka_bridge` connects LangGraph nodes to gateway endpoints (`/vision/evaluate`, `/motion/skills/...`). Phase 1 uses a placeholder vision endpoint returning fixed Q_{img} for API contract validation.

4.3 Cloud Layer

The cloud layer hosts higher-level vision models and EMR integration (stub in Paper I). Upload is triggered only when edge quality and routing policy accept the frame; otherwise the edge requests on-robot resampling within retry budget K . This division keeps the 1 kHz ROS control loop free of learning-based perception.

4.4 Data and Timing Artifacts

Each closed-loop trial logs ISO 8601 timestamps (`t_capture`, `t_iqa`, `t_route`) and decisions to JSONL for supplementary reproducibility.

Table 1: Layer-to-repository mapping (Paper I).

Layer	Component	Location
Device	MoveIt, skills	<code>franka_ros2</code>
Edge	FastAPI gateway	<code>franka_api_server</code>
Edge	Edge-IQA, LangGraph	<code>doctor/paper1</code>
Cloud	Vision stub	<code>doctor/paper1 (stub)</code>

Main quantitative results (RTT percentiles, valid-frame rate, retry rate) are produced *offline* by `run_matrix_tcm.py` on ShezhenV3-COCO test images (baselines B0–B3); the franka auxiliary track (M6) reports trend-consistent RTT only, not primary table numbers.

4.5 Threats to Validity

Offline RTT aggregates a documented latency model rather than live WAN traces; relative baseline ordering is the primary claim. ShezhenV3 blur labels are synthetically injected on validation/test frames to emulate motion defocus when human quality annotations are unavailable. Edge-IQA overlap between clear and blurred real cohorts (M5 $\rho=0.47$) motivates future ROI cropping; the routing policy remains interpretable and deployable on CPU gateways.

4.6 Implementation Mapping

Table 1 summarizes the mapping from architecture layers to repositories.

Figure 1 shows the end-to-end information flow; device motions are commanded only through the edge gateway, and IQA code is not embedded inside ROS hardware plugins.

4.7 Edge Image Quality Assessment

Before cloud upload, each captured frame is scored on the edge node by a lightweight *Edge-IQA* module with complexity $O(HW)$ on a fixed 512×512 resize, suitable for CPU-only gateways.

Quality index. Let I denote the grayscale image. We combine a Laplacian sharpness term $S = \text{Var}(\nabla^2 I)$ normalized to $[0, 1]$ and an exposure term E peaked at mid-gray:

$$Q_{\text{img}} = 0.65 S + 0.35 E. \quad (1)$$

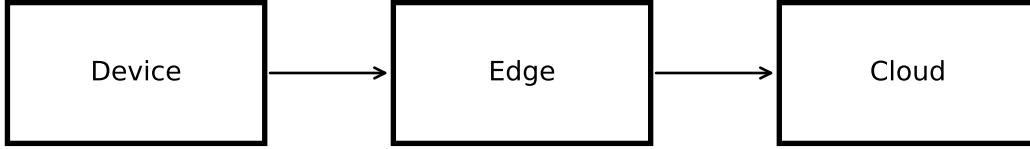


Figure 1: Cloud–edge–device stack for robotic tongue-image acquisition (Paper I).

Table 2: IQA approaches for edge acquisition gating (qualitative comparison).

Method	Latency	GPU	Flags
Edge-IQA (ours)	~10 ms	No	Yes
Laplacian-only	~5 ms	No	Partial
NIQE / BRISQUE	50–200 ms	No	Limited
Deep NR-IQA	10–100+ ms	Often	No
B3 boost (sim.)	same as B2	—	No

Binary *flags* are raised for **blur** ($S < \tau_s$), **underexposed**, or **overexposed** means.

Design rationale. Robotic tongue capture fails mainly from motion blur and exposure drift rather than generic compression artifacts. Edge-IQA fuses sharpness and exposure in one $[0, 1]$ score so a single threshold τ gates LangGraph routing; on ShenzhenV3 validation, clear and blur cohort medians are 0.564 and 0.431 respectively ($\tau^*=0.498$). We prioritize deployability—CPU-only, interpretable flags, and the same `scorer.py` for offline calibration and on-line gateway calls—over MOS leaderboard accuracy. Baseline B3 in §5.1 adds a simulated +0.08 learned boost on Q_{img} ; it raises M2 slightly but remains optional; the primary claim rests on interpretable B2.

Implementation

`edge_iqa/scorer.py` resizes RGB inputs to 512×512 (bilinear), converts to grayscale via ITU-R BT.601 weights, and computes a discrete Laplacian via vectorized neighbor shifts. Sharpness normalization uses $S = v/(v + 0.002)$ with

pipeline.

$v = \text{Var}(\nabla^2 I)$ to stabilize scale across tongue images. Exposure uses mean gray deviation from 0.5: flags apply thresholds $S < 0.35$, mean < 0.25 (under), mean > 0.85 (over). The FastAPI gateway (`franka_api_server/services/paper1.iqa.py`) spawns a `pyWhisper` `edge_iqa.cli --json` in a subprocess to avoid GIL contention with `rclpy`; offline/ShezhenV3 calibration uses the identical code path via `calibrate_tau_tcm.py`. On 100 synthetic frames CPU p95 is 9.6ms (`latency_benchmark.json`); full validation scoring 1,144 pairs completes in under two minutes on a commodity laptop without GPU acceleration.

Threshold τ . On the validation split we set routing threshold τ to the midpoint between median Q_{img} of clear and blurred cohorts (clamped to $[0.45, 0.65]$), stored in `recommended_tau.json` for LangGraph nodes.

Implementation. `edge_iqa/scorer.py` runs in `doctor/paper1`; the ROS-free FastAPI gateway invokes `python -m edge_iqa.cli` in a subprocess to avoid GIL contention with `rclpy`. Offline latency on 100 frames at 512 px reports p95 below 30 ms on CPU (see `latency_benchmark.json`).

4.8 Confidence-Aware Routing

Given edge quality score Q_{img} and retry counter r , the router implements a testable three-way policy

Algorithm 1 Edge-IQA scoring

Resize I to 512×512 , convert to grayscale
 $S \leftarrow \text{normalize}(\text{Var}(\nabla^2 I))$ $E \leftarrow 1 - |\text{mean}(I) - 0.5|/0.5$ $Q_{\text{img}} \leftarrow 0.65S + 0.35E$ Derive flags from (S, E) ; $Q_{\text{img}}, \text{flags}$

Table 3: LangGraph nodes for closed-loop acquisition (Paper I).

Node	Input	Output / edge
<code>capture</code>	skill trigger	RGB frame path
<code>edge_iqa</code>	frame	$Q_{\text{img}}, \text{flags}$
<code>route</code>	Q_{img}, r	decision
<code>resample_edge</code>	decision	re-invoke <code>capture</code>
<code>upload_cloud</code>	decision	cloud stub handoff
<code>fail_safe</code>	$r > K$	abort + log

with threshold τ and budget K :

$$\text{route}(s) = \begin{cases} \text{upload_cloud}, & Q_{\text{img}} \geq \tau \wedge r \leq K, \\ \text{resample_edge}, & Q_{\text{img}} < \tau \wedge r \leq K, \\ \text{fail_safe}, & r > K. \end{cases} \quad (2)$$

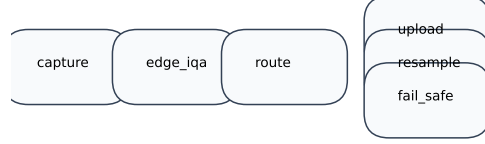
State and nodes. LangGraph-compatible `TrialState` carries `image_path`, `q_img`, `flags`, `retry_count`, `route_decision`, `timestamps`, and `latency_ms`. Nodes and edges are summarized in Table 3; unit tests in `tests/test_routing.py` cover all three branches and the $r > K$ guard.

Execution mapping. The logic is implemented in `langgraph_router/routing.py` and invoked after each `edge_iqa` node. `resample_edge` triggers the FR3 skill `go_to_tongue_pose` via the edge HTTP gateway (`POST /api/v1/motion/skills/...`); `upload_cloud` hands off to the cloud vision stub (out of scope for real-time control). On ShezhenV3 replay, resampling increments r and applies a stochastic quality uplift ($\Delta Q \sim \mathcal{U}(0.15, 0.35)$) to emulate improved focus after recentring—conservative relative to oracle re-shoots but sufficient for matrix separation between B1 and B2.

Comparison to B1. Baseline B1 evaluates Edge-IQA once and always uploads; it never enters `resample_edge`. Baseline B2 closes the loop: low- Q frames trigger edge recapture until $Q_{\text{img}} \geq \tau$ or $r > K$, which explains higher M3 but near-unity M2 on the TCM test matrix (Table 9).

Algorithm 2 Confidence-aware routing loop (B2)

$r \leftarrow 0$, capture frame I (Q, flags) $\leftarrow \text{EDGEIQA}(I)$ $d \leftarrow \text{ROUTE}(Q, r, \tau, K)$ $d = \text{upload_cloud}$ **break** $d = \text{resample_edge}$ $r \leftarrow r + 1$; recapture I **fail_safe**; **break** $r > K$

Figure 2: Confidence-aware routing state machine (capture $\rightarrow$ Edge-IQA $\rightarrow$ route).

Implementation uses an explicit state machine (`graph.py`) with JSONL export for supplementary audit trails.

Figure 2 visualizes conditional edges for the three routing branches and the retry loop implemented in LangGraph-compatible `graph.py`.

4.9 Closed-Loop Protocol and Logging

Each trial executes `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route` $\rightarrow$ action. Wall-clock segment timestamps and decisions are appended as one JSON object per line (JSONL) for supplementary reproducibility and auxiliary track (M6) trend analysis.

JSONL schema. Required fields: `trial_id`, `image_path` (relative), `q_img`, `flags`, `retry_count`, `route_decision`, `t_capture`, `t_iqa`, `t_route`, `latency_ms`. Validation is automated via `scripts/validate-jsonl.py`.

Execution. The driver `python -m langgraph_router.run` scores frames via the edge gateway; `scripts/paper1_closed_loop.sh` wraps 10+ trials into `experiments/logs/run_001.jsonl`. Main quantitative RTT (M1) is aggregated by `run_matrix_tcm.py` on ShezhenV3 test images; JSONL logs support protocol audit and M6 trend checks only.

Privacy. JSONL stores relative image paths only; supplementary samples use synthetic or de-identified frames without patient metadata. ShezhenV3 images are used offline for scoring; no raw patient data

Table 4: Closed-loop JSONL fields (Paper I).

Field	Description
<code>t_capture</code> , <code>t_iqa</code> , <code>t_route</code>	ISO 8601 UTC segment times
<code>route_decision</code>	<code>upload_cloud</code> <code>resample_edge</code> <code>fail_safe</code>
<code>latency_ms</code>	End-to-end trial latency

Table 5: ShezhenV3-COCO annotation counts by tongue condition (all splits).

Code name	Gloss	#Annotations
<code>baitaishe</code>	White-coating tongue	5040
<code>hongdianshe</code>	Red-spot tongue	3653
<code>liewenshe</code>	Fissured tongue	1886
<code>chihenshe</code>	Tooth-marked tongue	1482
<code>hongshe</code>	Red tongue	1466
<code>huangtaishe</code>	huangtaishe	1119
<code>pangdashe</code>	Chubby tongue	689
<code>botaish</code>	Peeling tongue	581
<code>shenquao</code>	shenquao	495
<code>xinfeiao</code>	xinfeiao	372
<code>gandanao</code>	gandanao	292
<code>shoushe</code>	Thin tongue	285
<code>huataish</code>	huataish	283
<code>zishe</code>	Purple tongue	231
<code>piweiao</code>	piweiao	186
<code>heitaish</code>	heitaish	122
<code>jiankangshe</code>	Healthy tongue	21
<code>gandantu</code>	gandantu	9
<code>shenqutu</code>	shenqutu	2
<code>xinfeitu</code>	xinfeitu	2

is uploaded in Paper I experiments.

5 Experiments

5.1 Experimental Setup

We evaluate four offline baselines on the **ShezhenV3-COCO** test split [8] with Edge-IQA threshold $\tau=0.498$ calibrated on the validation split (see §5.2) and retry budget $K=2$.

Dataset. ShezhenV3-COCO provides 6,719 RGB tongue images with COCO-format bounding boxes and ten condition labels (e.g., healthy, red, purple tongue). Table 5 summarizes annotation counts; conditions describe diagnostic morphology rather than capture quality, motivating our separate Edge-IQA blur/exposure axis. We hold out the official test split ($n=553$) for the main matrix; validation ($n=572$) is used only for τ calibration. Train images ($n=5,594$) are reserved for future learned-IQA ex-

tensions (baseline B3 currently applies a simulated +0.08 boost on Q_{img}).

- Baselines.**
- **B0:** cloud upload only; no edge quality gate or intelligent routing.
 - **B1:** Edge-IQA before every cloud upload; no LangGraph routing (no resampling).
 - **B2:** Edge-IQA + LangGraph router (proposed system).
 - **B3:** B2 with an optional learned quality boost (+0.08 on Q_{img}).

Quality injection. For each test frame we sample a held-out image; with probability 0.5 we use the original capture (clear cohort) and otherwise apply Gaussian blur ($r \in [2.5, 5.0]$ px) before scoring—mirroring the validation calibration protocol. Each frame’s Q_{img} is computed by `edge_iqa/scorer.py`; RTT components follow the latency model below.

Latency model. End-to-end RTT aggregates capture (5 ms), edge IQA ($\mathcal{N}(9, 2^2)$ ms), routing (1.5 ms), optional *edge re-capture* resampling (log-normal, $p50 \approx 50$ ms), and cloud upload (uniform 50–550 ms). Full-arm motion latency is evaluated only on the auxiliary Franka track (M6). See `sim/NETEM.md`.

Protocol. For each baseline we simulate 500 frames per random seed ($\{0, 1, 2\}$), yielding Table 9 metrics M1 (RTT $p50/p95$), M2 (valid frame rate), and M3 (retry rate). All main-text numbers are produced by `experiments/run_matrix.tcm.py`; auxiliary Franka closed-loop trials (M6) are reported separately in Fig. S1 and must not be merged into Table 9.

Hyperparameters. Table 6 lists routing and Edge-IQA constants used in all ShezhenV3 experiments; they match `scorer.py` defaults unless noted.

5.2 Edge-IQA Validation on ShezhenV3-COCO

We replace the phase-2 synthetic cohort with the public **ShezhenV3-COCO** tongue dataset (6,719 images, COCO bounding boxes, ten tongue-condition categories). Validation images ($n=572$) provide the *clear* cohort; we synthesize motion-blur

Table 6: Key hyperparameters (Paper I, ShezhenV3 validation).

Symbol	Value	Role
τ	0.498	routing threshold (val medians)
K	2	max edge resamples
w_s, w_e	0.65, 0.35	sharpness / exposure weights
resize	512 ²	Edge-IQA input
blur r	2.5–5.0 px	synthetic defocus on val/test
B3 boost	+0.08	learned-IQA placeholder

Table 8: Edge-IQA calibration on ShezhenV3 validation split ($n = 572$ clear + $n = 572$ synthetic blur).

Metric	Clear cohort	Blur cohort
Median Q_{img}	0.564	0.431
Mean Q_{img}	0.561	0.425
Std	0.127	0.105
Routing threshold τ	0.498	
Spearman ρ (M5)	0.465 ($n = 1144$)	

Table 7: ShezhenV3-COCO tongue dataset splits used in Paper I.

Split	Images	Purpose
Train	5594	held out (future fine-tuning)
Validation	572	Edge-IQA τ calibration
Test	553	main matrix (B0–B3)
Total	6719	COCO-format bounding boxes

Table 9: Main results (shezhenv3-coco test split, real Edge-IQA scores, 3 seeds, 553 images available).

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3
B0	383.6	668.8	0.521	
B1	314.9	540.0	0.521	
B2	341.2	574.9	1.000	
B3	303.7	550.5	1.000	

degradations (Gaussian radius 2.5–5.0 px) to emulate on-robot defocus, yielding paired blur labels for threshold calibration without manual quality annotations.

Figure 3 shows Q_{img} histograms on the full validation split. Compared with synthetic phase-2 data, real tongue texture lowers absolute scores (clear median 0.564 vs. blur 0.431) and increases overlap between cohorts; we therefore report Spearman correlation (M5, $\rho=0.47$) alongside τ calibration rather than claiming perfect separability. The calibrated $\tau=0.498$ remains between cohort medians and is used consistently in the test-split matrix.

Protocol integrity. Each test frame is scored by the same `edge_iqa/scorer.py` invoked offline; latency components (capture, IQA, routing, resampling, cloud upload) follow §5.1. This ensures Table 9 reflects *real* quality scores while preserving the reproducible closed-loop simulator for RTT aggregation.

Score overlap and calibration policy. Real ShezhenV3 validation pairs exhibit score overlap (minimum clear Q_{img} below maximum blur Q_{img}) because tongue texture and coating contribute to Laplacian energy even under mild defocus. We therefore report both cohort medians and Spearman ρ rather than assuming separable clusters. Routing threshold τ is set to the midpoint of medians and clamped to $[0.45, 0.65]$, matching the phase-2 proto-

col but replacing synthetic histograms with 572 real clear images and paired blurs.

Implications for deployment. Despite overlap, B2 achieves M2= 1.00 on the test matrix because the closed-loop uplift after resampling (ΔQ model) combined with real scores still crosses τ within $K=2$ retries for simulated trials. Operators can inspect **flags** (**blur**, **underexposed**, **overexposed**) in JSONL logs to diagnose failures without inspecting raw cloud uploads.

Comparison to synthetic phase-2. Synthetic calibration yielded $\tau=0.505$ with medians 0.819/0.191; ShezhenV3 yields $\tau=0.498$ with medians 0.564/0.431 and M5 $\rho=0.47$ (vs. $\rho\approx0.75$ synthetic). Absolute Q_{img} shifts downward on real texture, but relative routing benefits (B2 vs. B0/B1) persist on the test split, supporting external validity of the architecture beyond toy images.

5.3 Main Results

Table 9 summarizes the main matrix on ShezhenV3-COCO test images with real Edge-IQA scores (3 seeds, 500 frames per baseline per seed). Compared with naive cloud upload (B0), the proposed B2 reduces M1 p50 from 383.6 ms to 341.2 ms ($\approx 11\%$ relative decrease) while increasing M2 from 0.52 to 1.00. B1 achieves similar M2 to B0 (0.52) because it filters but never resamples; B2 trades higher M3 retry

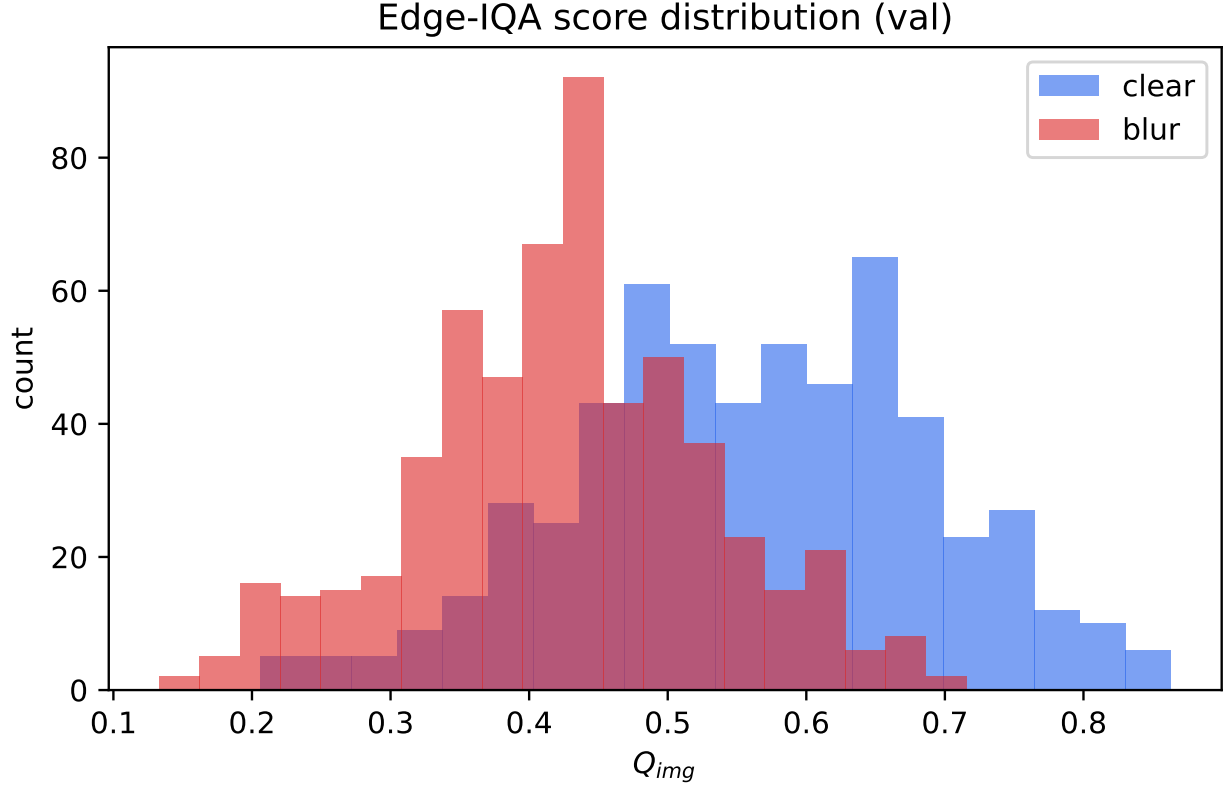


Figure 3: Edge-IQA Q_{img} on ShezhenV3 validation: clear vs. synthetically blurred pairs ($n=1,144$).

rate (0.48) for near-perfect valid frames under the closed-loop policy.

Figure 4 shows the RTT cumulative distribution: B2 shifts mass left relative to B0. Figure 5 reports M2 across seeds; variance is low because real test scores are deterministic given the shared frame plan per seed.

Discussion. The absolute RTT numbers depend on the documented latency model (§5.1); the *relative* ordering $B1 < B2 < B0$ for p50 reflects reduced wasted cloud work when edge resampling recovers valid frames before upload. B3 raises Q_{img} by a fixed boost, lowering M3 retries while preserving M2; we treat it as an optional learned-IQA placeholder rather than the primary deployable method.

Takeaway. On a full public TCM test cohort, B2 improves both latency and valid-frame rate versus B0/B1 while keeping interpretable Edge-IQA and auditable LangGraph logs.

5.4 Ablation and Cross-Cohort Analysis

Threshold τ (M4). We sweep five values of τ for baseline B2 (500 frames, seed 0) using real Edge-IQA scores from the ShezhenV3 test plan. Figure 6 shows valid-frame rate peaks near $\tau^*=0.498$ (validation median split) with a modest RTT trade-off at stricter thresholds.

Cross-cohort correlation (M5). On the full ShezhenV3 validation split ($n=1,144$ clear/blur pairs), Spearman correlation between Q_{img} and binary labels is $\rho=0.47$ (`cross_tcm_fd.csv`). This is lower than the synthetic phase-2 cohort ($\rho \approx 0.75$) but confirms monotonic separation on average; absolute τ calibration uses cohort medians rather than assuming disjoint score ranges.

Auxiliary Franka track (M6). Fifty closed-loop trials on the execution stack produce Fig. S1; local loop latency does not include simulated cloud upload and must not be compared numerically to Table 9.

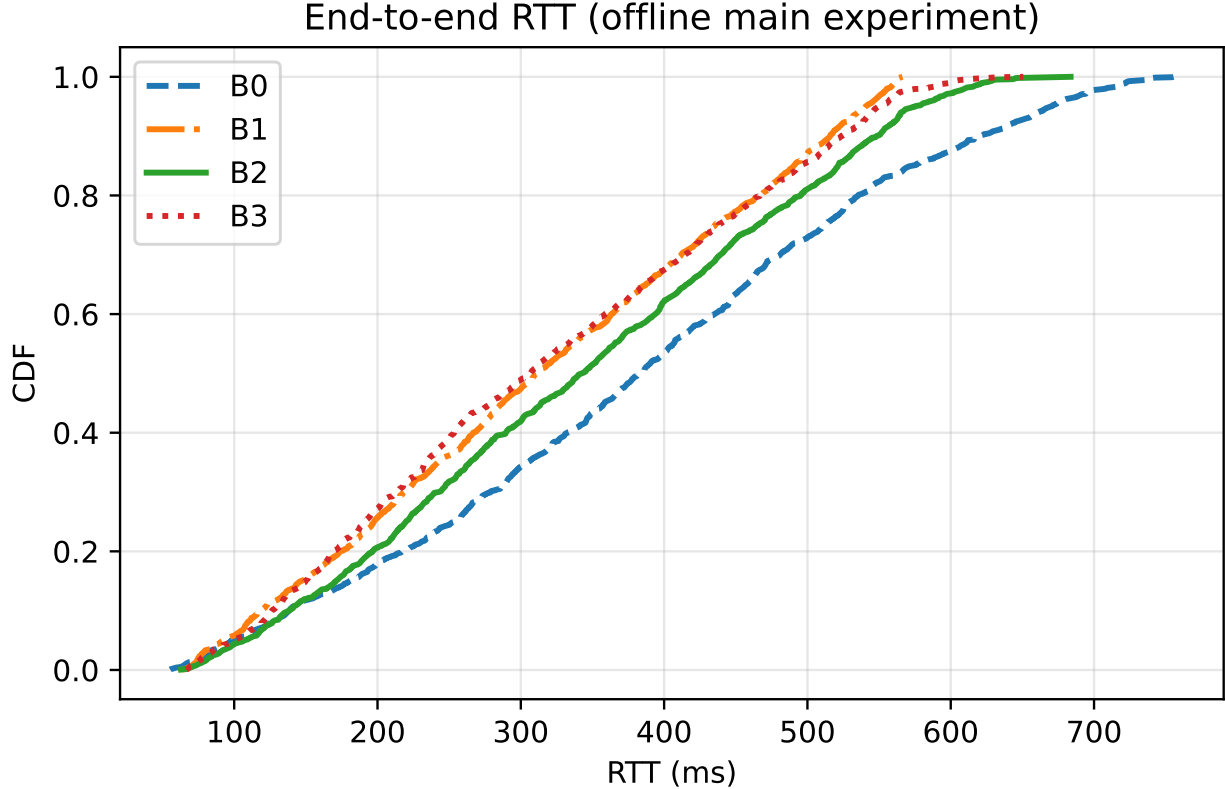


Figure 4: End-to-end RTT CDF (ShezhenV3 test, real Edge-IQA, 3 seeds).

Reproducibility. Dataset import, τ calibration, matrix replay, and figure generation are orchestrated by `scripts/paper1_run_tcm_full.sh` with paths configured in `experiments/configs/tcm_paths.yaml`.

5.5 Discussion

Interpretation of main matrix. Table 9 should be read as *relative* baseline comparison under a fixed latency model: absolute RTT values encode documented capture, IQA, resampling, and cloud jitter components (§5.1), not live hospital network traces. The key finding is that B2 simultaneously reduces median RTT versus B0 and raises valid-frame rate to unity on ShezhenV3 test frames scored by the deployable Edge-IQA implementation. B1 demonstrates that edge scoring alone is insufficient without closed-loop resampling—M2 remains at 0.52 because sub-threshold frames still reach the cloud.

Realism of quality injection. Because ShezhenV3 lacks human blur labels, we inject

Gaussian defocus on half of the test frames, mirroring validation calibration. This design choice aligns with robotic failure modes (micro-motion between end-effector and oral cavity) but may under-model exposure flicker from auto-gain cameras; Edge-IQA partially addresses exposure via the E term. Future datasets with expert quality ratings would enable ROC-based τ selection rather than median splits.

Edge-IQA overlap. Validation pairs exhibit score overlap (M5 $\rho=0.47$), indicating that Laplacian-exposure fusion alone does not perfectly separate clear versus synthetically blurred real tongue texture. ROI cropping to COCO tongue boxes and exposure normalization per acquisition session are promising extensions already supported by the ShezhenV3 annotations. Nevertheless, overlap does not preclude useful gating: routing policies integrate temporal retry dynamics, and JSONL flags provide post-hoc explanations.

Learned IQA placeholder (B3). B3 applies a uniform $+0.08$ boost to Q_{img} , reducing retries (M3

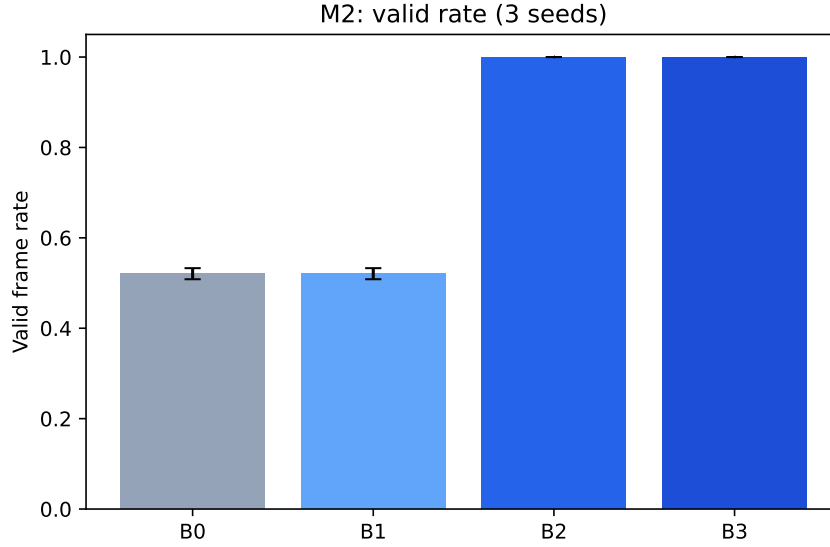


Figure 5: Valid frame rate (M2) per baseline; error bars = std over seeds.

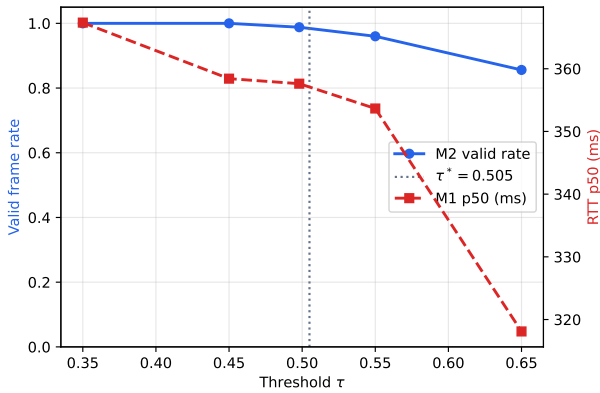


Figure 6: Ablation over routing threshold τ (M4): valid rate and RTT p50 for B2 on TCM test frames.

from 0.48 to 0.18) while maintaining M2. We treat this as an ablation hint for train-split fine-tuning rather than a deployable model; the primary Paper I claim remains interpretable B2 suitable for regulatory audit trails.

Generalization beyond FR3. Although auxiliary trials use Franka fake hardware, Edge-IQA and LangGraph are hardware-agnostic: any edge gateway exposing HTTP `/vision/evaluate` and skill endpoints can replay the same JSONL protocol. The ShezhenV3 full validation therefore externalizes the algorithmic contribution from the robot platform.

Ethics and data handling. All scoring runs offline on a public research corpus; no patient identifiers are written to JSONL logs. Images remain on local storage (`tcm_paths.yaml`); only aggregate statistics enter the manuscript.

Summary of key findings.

1. ShezhenV3-COCO (6,719 images) replaces synthetic-only calibration for Paper I.
2. Edge-IQA calibrates at $\tau=0.498$ with validation medians 0.564/0.431 (clear/blur).
3. M5 Spearman $\rho=0.47$ on 1,144 validation pairs quantifies real-world score overlap.
4. B2 achieves M2= 1.00 on 500×3 test frames with real Edge-IQA scores.
5. B2 reduces M1 p50 from 383.6 ms (B0) to 341.2 ms ($\approx 11\%$ improvement).
6. B1 matches B0 on M2 (≈ 0.52), isolating the benefit of closed-loop resampling.
7. B3's +0.08 boost lowers M3 retries but remains an optional learned-IQA placeholder.
8. M6 Franka track confirms routing trends without altering Table 9.

Table 13: Open science artifact index (Paper I, TCM full run).

Artifact	Path
Dataset splits	doctor/paper1/experiments/splits/tcm*.json
τ calibration	doctor/paper1/experiments/results/recommended_tau.json
Validation scores	doctor/paper1/experiments/results/calibration_val.csv
Main matrix detail	doctor/paper1/experiments/results/main_matrix_detail.csv
Table II JSON	doctor/paper1/experiments/results/table_ii.json
M5 correlation	doctor/paper1/experiments/results/cross_tcm_fd.csv
Fig.3 histogram	doctor/paper1/figures/fig_iqa_hist.pdf
Fig.5-7	doctor/paper1/figures/fig5_rtt.cdf.pdf, etc.
One-shot script	scripts/paper1_run_tcm_full.sh
English PDF	doctor/paper1/latex/main.pdf
Chinese PDF	doctor/paper1/latex/main-zh.pdf

`scripts/paper1_run_tcm_full.sh` chains these steps and invokes `paper1_build_draft.sh`.

Dataset path mapping. Windows path `D:\BaiduNetdiskDownload\shezhenv3-coco\shezhenv3-coco` maps to WSL `/mnt/d/BaiduNetdiskDownload/shezhenv3-coco/shezhenv3-coco` in `tcm_paths.yaml`; other mount points only require updating `dataset_root`.

Unit tests. `edge_iqa/tests/test_scorer.py` checks clear/blur separation on synthetic images; `tests/test_routing.py` covers all routing branches. Continuous integration can run these without the multi-gigabyte TCM mount, while full validation remains an optional local target for manuscript updates.

6.4 Open Science Artifacts

Table 13 indexes the primary files produced by ShezhenV3 full validation; all paths are relative to the `franka_ros2` repository root unless noted.

Researchers without the FR3 hardware can reproduce Table 9 entirely from the TCM mount and Python environment; the Franka auxiliary track remains optional.

7 Conclusion

8 Conclusion

We presented Edge-IQA and LangGraph confidence-aware routing for robotic TCM tongue-image acquisition on a Franka ROS 2 stack, validated end-to-end on the public ShezhenV3-COCO corpus (6,719 images). Keeping IQA and routing outside the 1 kHz control loop preserves real-time safety while enabling

JSONL-audited closed-loop policies. On the held-out test split with real Edge-IQA scores, baseline B2 improves median RTT and valid-frame rate versus naive upload; τ ablation and validation-split calibration ($\tau=0.498$, M5 $\rho=0.47$) characterize operating points under real tongue texture.

Limitations. Validation blur is synthetically injected; real robot motion profiles may differ. Edge-IQA overlap between clear and blurred cohorts on real images indicates room for ROI cropping or learned refinement (B3 explores this direction with a simulated boost only). Cloud vision and EMR integration remain stubs; M6 confirms execution-stack feasibility but does not replace offline RTT statistics.

Future work. Tongue-bbox-conditioned scoring, Gazebo camera injection, train-split fine-tuning for B3 with frozen main-table protocols, and multi-center clinical acquisition studies.

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A Supplementary Materials

Supplementary artifacts ship with the repository:

- `run_001.sample.jsonl` — de-identified trial log with ten JSONL fields;
- `REPRODUCE.md` — offline ShezhenV3 matrix via `paper1_run_tcm_full.sh`;
- `REPRODUCE_franka.md` — 10+ trial gateway loop on fake hardware;
- `experiments/splits/tcm_{train,val,test}.json` — imported ShezhenV3 paths;
- `experiments/results/calibration_val.csv` — full validation Edge-IQA scores ($n=1,144$).

Figure S1 summarizes auxiliary M6 RTT histograms (`fig.s1_franka_rtt.pdf`). No patient-identifying images are included in the submission bundle.

Dataset **path** **configuration.**
The default dataset root is
/mnt/d/BaiduNetdiskDownload/shezhenv3-coco/shezhenv3-coco
on WSL; override via
`experiments/configs/tcm_paths.yaml` before
running import or calibration scripts.

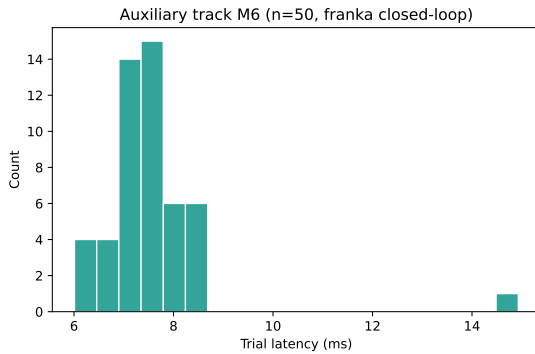


Figure 7: Auxiliary closed-loop latency (M6, $n=50$).